LEARNING HUB MASTER SERIES — LINEAR ALGEBRA

Positive Definite Matrices & Quadratic Forms

Energy Topography, The 5 Equivalence Criteria, Cholesky Factorization, Ellipsoid Principal Geometry, and Rayleigh Optimization

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Core Curriculum Grounding:

- Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapters 6.5 & 7.1–7.2)
- Kenji Hiranabe — *The Art of Linear Algebra* (Matrix World, Spectral Projections, and Ellipsoid Maps)
- MIT 18.06 & Advanced University Engineering Linear Algebra Reference Series

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Part I: Pure Quadratic Forms & Energy Topography

1.1 From Linear Systems to Quadratic Energy: The Bilinear Form f(x) = xᵀ S x

In foundational linear algebra, our primary focus rests upon linear equations of the form $S\mathbf{x} = \mathbf{b}$, representing flat hyperplanes intersecting in vector spaces. However, modern engineering, physics, machine learning, and optimization invariably demand the minimization of quadratic functions representing kinetic, potential, electrical, or structural energy.

A **pure quadratic form** in n variables is a homogeneous polynomial of degree two:

$$f(\mathbf{x}) = \mathbf{x}^T S\,\mathbf{x} = \sum_{i=1}^{n} \sum_{j=1}^{n} s_{ij}\, x_i\, x_j$$

Theorem 1.1 (The Symmetric Representation Principle)

Any real quadratic form $f(\mathbf{x}) = \mathbf{x}^T A\,\mathbf{x}$ generated by an arbitrary square matrix A can be represented **uniquely** by a real symmetric matrix $S = \frac{1}{2}(A + A^T)$. The skew-symmetric component $K = \frac{1}{2}(A - A^T)$ contributes identically zero energy for all $\mathbf{x} \in \mathbb{R}^n$:

$$\mathbf{x}^T K\,\mathbf{x} = (\mathbf{x}^T K\,\mathbf{x})^T = \mathbf{x}^T K^T \mathbf{x} = -\mathbf{x}^T K\,\mathbf{x} \Rightarrow 2\,\mathbf{x}^T K\,\mathbf{x} = 0 \Rightarrow \mathbf{x}^T K\,\mathbf{x} = 0$$

Consequently, throughout quadratic form theory and definiteness analysis, we assume **without loss of generality** that the coefficient matrix S is real and symmetric ($S = S^T$).

In the two-variable setting $\mathbf{x} = [x_1, x_2]^T$, the expansion produces:

$$f(x_1, x_2) = \begin{vmatrix} x_1 & x_2 \end{vmatrix} \begin{vmatrix} a & b \\ b & c \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = a\,x_1^2 + 2b\,x_1\,x_2 + c\,x_2^2$$

Notice that the diagonal entries a and c govern the pure square terms x_1^2 and x_2^2, while the off-diagonal entry b is shared equally across the cross-product term $2bx_1x_2$.

In three variables $\mathbf{x} = [x_1, x_2, x_3]^T$, the general quadratic form expands into:

$$f(x_1, x_2, x_3) = s_{11}\,x_1^2 + s_{22}\,x_2^2 + s_{33}\,x_3^2 + 2s_{12}\,x_1\,x_2 + 2s_{13}\,x_1\,x_3 + 2s_{23}\,x_2\,x_3$$

1.2 Completing the Square & Gaussian Elimination: The S = LDLᵀ Pivot Bridge

The central analytical challenge in analyzing $f(\mathbf{x})$ is the cross-product terms (e.g., $2bx_1x_2$), which entangle the coordinate variables. In elementary algebra, we eliminate cross-terms by **completing the square**. In linear algebra, completing the square is identical to **Gaussian elimination** factoring $S = L\,D\,L^T$!

Derivation 1.2 (The Elimination-Completion Equivalence)

Consider $f(x_1, x_2) = a\,x_1^2 + 2b\,x_1\,x_2 + c\,x_2^2$ with $a \neq 0$:

$$f(x_1, x_2) = a \cdot \underbrace{[x_1^2 + 2(\frac{b}{a})\,x_1\,x_2 + (\frac{b}{a})^2 x_2^2]}_{} + \underbrace{(c - \frac{b^2}{a})}_{}\,x_2^2 = a\,\underbrace{(x_1 + \frac{b}{a}\,x_2)^2}_{} + \underbrace{(c - \frac{b^2}{a})}_{}\,x_2^2$$

Now examine the symmetric Gaussian elimination factorization $S = L\,D\,L^{\mathsf{T}}$:

$$\begin{vmatrix} a & b \\ b & c \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ b/a & 1 \end{vmatrix} \begin{vmatrix} a & 0 \\ 0 & c - b^2/a \end{vmatrix} \begin{vmatrix} 1 & b/a \\ 0 & 1 \end{vmatrix} = L\,D\,L^{\mathsf{T}}$$

The mathematical identity is complete: The coefficients outside the squared parentheses are **precisely the elimination pivots** $d_1 = a$ and $d_2 = c - b^2/a$! The multiplier inside the first squared term is the elimination entry $l_{21} = b/a$.

In general n-dimensional space, Gaussian elimination without row exchanges expresses any symmetric matrix as $S = L\,D\,L^{\mathsf{T}}$. Setting $\mathbf{y} = L^{\mathsf{T}}\,\mathbf{x}$ (a triangular coordinate transformation), the quadratic form decouples into a weighted sum of independent squares:

$$f(\mathbf{x}) = \mathbf{x}^{\mathsf{T}} S\,\mathbf{x} = \mathbf{x}^{\mathsf{T}} (L\,D\,L^{\mathsf{T}})\,\mathbf{x} = (L^{\mathsf{T}}\,\mathbf{x})^{\mathsf{T}}\,D\,(L^{\mathsf{T}}\,\mathbf{x}) = \mathbf{y}^{\mathsf{T}}\,D\,\mathbf{y} = \sum_{i=1}^{n} d_i\,y_i^2$$

Because each $y_i^2 \geq 0$, the overall algebraic sign of the energy $f(\mathbf{x})$ is dictated **exclusively by the signs of the elimination pivots** $d_1, d_2, \ldots, d_n$!

1.3 Topographical Classification: Energy Bowl, Inverted Dome, Saddle Surface & Trough

The signs of the pivots (and equivalently, the signs of the eigenvalues) dictate the geometry of the energy surface $z = f(x_1, x_2)$ in $\mathbb{R}^3$ around the stationary point (0, 0):

- **Positive Definite ($S \succ 0$):** All pivots $d_i > 0$ (all $\lambda_i > 0$). The surface is an upward-opening **paraboloid bowl**. The energy $f(\mathbf{x}) > 0$ for every non-zero vector, reaching a strict absolute minimum of zero at the origin.
- **Negative Definite ($S \prec 0$):** All pivots $d_i < 0$ (all $\lambda_i < 0$). The surface is a downward-opening **inverted dome**. The origin is a strict absolute maximum.
- **Indefinite:** Pivots have mixed signs (at least one positive and one negative). The surface is a **hyperbolic saddle point** (Pringles chip). The origin is neither a minimum nor a maximum.
- **Positive Semidefinite ($S \succeq 0$):** Pivots satisfy $d_i \geq 0$ with at least one pivot equal to zero. The surface is a **parabolic trough (flat valley)**, where energy is zero along the entire nullspace of S.

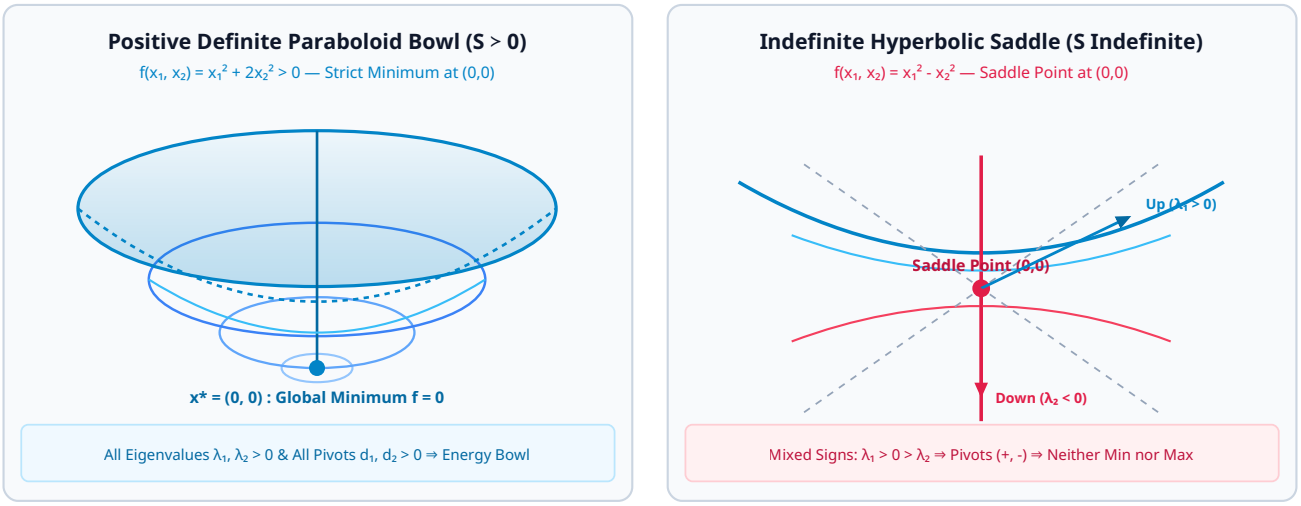


Figure 8.1: 3D Topography of Quadratic Forms — Convex Energy Bowl ($S \succ 0$) vs. Indefinite Hyperbolic Saddle Surface.

Part II: The 5 Equivalent Tests for Positive Definiteness

2.1 The Grand Equivalence Theorem of Positive Definiteness

Positive definiteness is one of the most powerful and unifying concepts in mathematics. Rather than having isolated definitions across algebra, analysis, and mechanics, positive definiteness admits five distinct characterizations that are completely equivalent.

Theorem 2.1 (The 5 Fundamental Tests for Positive Definiteness)

Let $S = S^T$ be a real symmetric $n \times n$ matrix. Then S is **positive definite** (denoted $S \succ 0$) if and only if it satisfies **any one** (and therefore all) of the following five conditions:

1. **Energy Test:** $\mathbf{x}^T S \mathbf{x} > 0$ for all non-zero vectors $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$.
2. **Eigenvalue Test:** All n eigenvalues are strictly positive: $\lambda_i > 0$ for $i = 1, \ldots, n$.
3. **Pivot Test:** All n pivots $d_1, d_2, \ldots, d_n$ in elimination without row exchanges are strictly positive.
4. **Sylvester's Criterion (Leading Principal Minors Test):** All n upper-left leading subdeterminants are strictly positive:

$$\det(S_k) > 0 \quad \text{for} \quad k = 1, 2, \ldots, n, \quad \text{where} \quad S_k = S(1{:}k, 1{:}k)$$

5. **Gram Matrix Factorization Test:** $S = A^T A$ for some real matrix A (size $m \times n$ with $m \geq n$) having linearly independent columns ($\text{rank}(A) = n$).

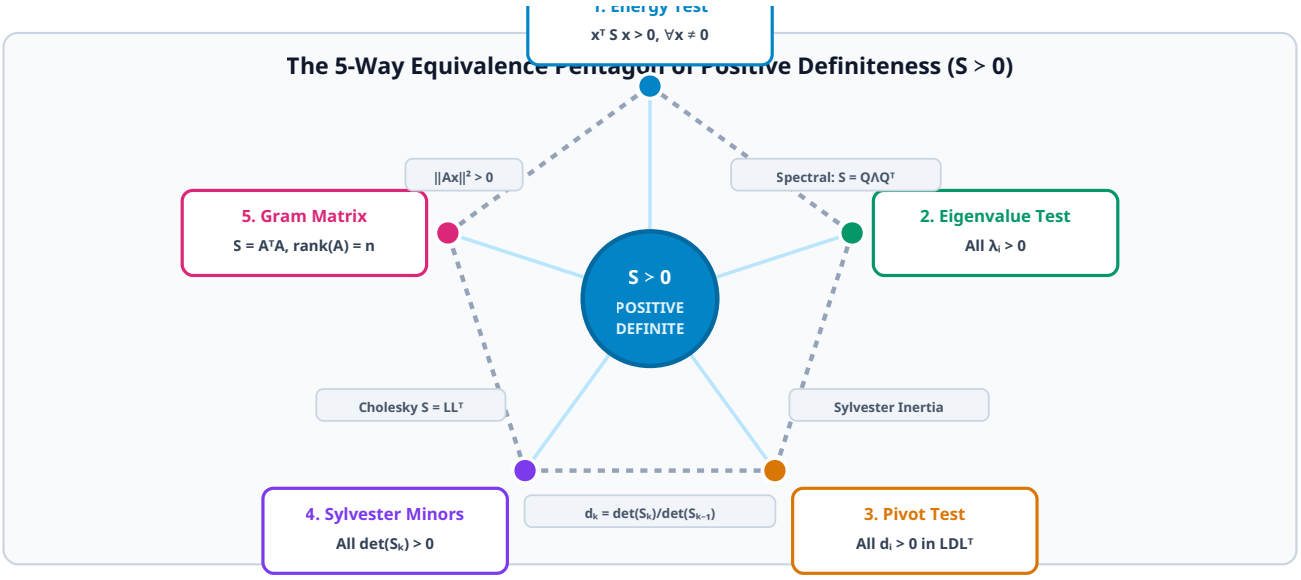


Figure 8.2: The 5-Way Equivalence Pentagon of Positive Definiteness — Mathematical Bridge Theorems Connecting All Five Tests.

2.2 Formal Mathematical Proof of the Equivalence Pentagon Cycle

To prove that all five conditions are mathematically equivalent, it suffices to establish a circular chain of implications alongside the Gramian factorization:

Proof of Equivalence: The Cycle (1) ⇒ (2) ⇒ (1) and (2) ⇔ (3) ⇔ (4) and (1) ⇔ (5)

Step 1: (1) ⇒ (2) [Energy ⇒ Eigenvalues].
Suppose $\mathbf{x}^T S \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. Let $(\lambda_i, \mathbf{q}_i)$ be any eigenpair with $\|\mathbf{q}_i\| = 1$. Then:

$$0 < \mathbf{q}_i^T S \mathbf{q}_i = \mathbf{q}_i^T (\lambda_i \mathbf{q}_i) = \lambda_i (\mathbf{q}_i^T \mathbf{q}_i) = \lambda_i (1) = \lambda_i$$

Hence every eigenvalue λ_i must be strictly positive.

Step 2: (2) ⇒ (1) [Eigenvalues ⇒ Energy].
By the Spectral Theorem (Volume 07), any symmetric S can be orthogonally diagonalized as $S = Q \Lambda Q^T$ with $Q^T Q = I$. For any $\mathbf{x} \neq \mathbf{0}$, define $\mathbf{y} = Q^T \mathbf{x}$. Since Q is invertible, $\mathbf{y} \neq \mathbf{0}$. Then:

$$\mathbf{x}^T S \mathbf{x} = \mathbf{x}^T (Q \Lambda Q^T) \mathbf{x} = (Q^T \mathbf{x})^T \Lambda (Q^T \mathbf{x}) = \mathbf{y}^T \Lambda \mathbf{y} = \sum_{i=1}^{n} \lambda_i y_i^2$$

Since all $\lambda_i > 0$ and at least one $y_i \neq 0$, the sum $\sum \lambda_i y_i^2 > 0$ strictly. Thus (1) and (2) are fully equivalent.

Step 3: (2) ⇔ (3) [Eigenvalues ⇔ Pivots via Sylvester's Inertia].
In Gaussian elimination without row exchanges, $S = L D L^T$ where L is unit lower triangular. This is a **congruence transformation** $D = L^{-1} S (L^{-1})^T$. By **Sylvester's Law of Inertia**, congruence transformations preserve the exact number of positive, negative, and zero eigenvalues. Because D is diagonal, its eigenvalues are precisely its diagonal entries (the pivots d_i). Hence, the number of positive eigenvalues of S equals the number of positive pivots. Thus all $\lambda_i > 0$ if and only if all $d_i > 0$.

Step 4: (3) ⇔ (4) [Pivots ⇔ Sylvester Leading Minors].
During triangular factorization of the upper-left $k \times k$ submatrix, $S_k = L_k D_k L_k^T$. Taking determinants:

$$\det(S_k) = \det(L_k) \det(D_k) \det(L_k^T) = (1) \cdot (d_1 d_2 \dots d_k) \cdot (1) = \prod_{j=1}^{k} d_j$$

This yields the fundamental quotient relationship:

$$d_1 = \det(S_1), \qquad d_k = \frac{\det(S_k)}{\det(S_{k-1})} \qquad (\text{for } k = 2, \dots, n)$$

By simple induction: all $d_k > 0$ if and only if all $\det(S_k) > 0$.

Step 5: (1) ⇔ (5) [Energy ⇔ Gram Matrix Factorization].
(⇒) Suppose $S = A^T A$ where A has linearly independent columns ($\text{rank}(A) = n$). Then for any $\mathbf{x} \neq \mathbf{0}$:

$$\mathbf{x}^T S \mathbf{x} = \mathbf{x}^T A^T A \mathbf{x} = (A\mathbf{x})^T (A\mathbf{x}) = \|A\mathbf{x}\|^2 \geq 0$$

Since $\text{rank}(A) = n$, the nullspace of A is trivial: $A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$. Thus for $\mathbf{x} \neq \mathbf{0}$, $\|A\mathbf{x}\|^2 > 0$ strictly. Therefore $S \succ 0$.
(⇐) Conversely, if $S \succ 0$, Test 3 guarantees $S = L D L^T$ with all $d_i > 0$. Since $D > 0$, its square root $D^{1/2} = \text{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})$ exists with real, positive diagonal entries. Setting $A = (L D^{1/2})^T = D^{1/2} L^T$, we have:

$$A^T A = (L D^{1/2})(D^{1/2} L^T) = L D L^T = S$$

Because A is square upper-triangular with non-zero diagonal entries $\sqrt{d_i}$, its columns are linearly independent. •

Part III: Positive Semidefinite Matrices & Cholesky Decomposition

3.1 Positive Semidefiniteness ($S \succeq 0$) & The Sylvester Minor Pitfall

A matrix is **positive semidefinite** (denoted $S \succeq 0$) if its energy satisfies $\mathbf{x}^T S \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$. This allows the energy to reach zero along non-zero vectors in the nullspace of S.

⚠️ **CRITICAL PITFALL: Leading Minors Fail for Semidefiniteness!**

Students frequently assume that testing $\det(S_k) \geq 0$ for all leading principal minors guarantees positive semidefiniteness. **This is mathematically FALSE!**

Counterexample: Consider the diagonal matrix $S = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$.

The leading principal minors are: $\det(S_1) = 0 \geq 0$ and $\det(S_2) = 0 \geq 0$. Both leading minors are non-negative! However, the eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = -1$. The matrix is **negative semidefinite**, not positive semidefinite! Along the vector $\mathbf{x} = [0, 1]^T$, the energy is $\mathbf{x}^T S \mathbf{x} = -1 < 0$!

The Correct Rule for $S \succeq 0$: To verify positive semidefiniteness via determinants, **ALL principal subdeterminants** (all $2^n - 1$ principal minors formed by choosing any subset of rows and corresponding columns), not just the leading upper-left minors, must be ≥ 0. For $n = 2$, this requires $s_{11} \geq 0$, $s_{22} \geq 0$, and $\det(S) \geq 0$.

3.2 Cholesky Factorization $S = LL^T$: Existence, Uniqueness, and Efficiency

For any symmetric positive definite matrix $S \succ 0$, the symmetric decomposition $S = L D L^T$ can be condensed into a single triangular factor. Because all pivots $d_i > 0$, we can absorb the square root of D directly into L:

$$S = L_0 D L_0^T = (L_0 D^{1/2})(D^{1/2} L_0^T) = L_{chol} L_{chol}^T$$

where $L_{chol} = L_0 D^{1/2}$ is a real lower-triangular matrix with strictly positive diagonal entries:

$$(L_{chol})_{ii} = \sqrt{d_i} > 0$$

Engineering Superiority of Cholesky Factorization

In scientific computing and data science, Cholesky factorization is the algorithm of choice for solving normal equations $(A^{\mathsf{T}}A)\mathbf{x} = A^{\mathsf{T}}\mathbf{b}$, Kalman filters, and Gaussian process regression for three compelling reasons:

- **Computational Cost:** Requires only $\frac{1}{3}n^3$ arithmetic operations — exactly **half the computational effort** of standard LU decomposition ($\frac{2}{3}n^3$).
- **Storage Efficiency:** Because $U = L^{\mathsf{T}}$, only the lower triangle of S needs to be stored or processed in memory.
- **Unconditional Numerical Stability:** No row exchanges or pivoting strategies (such as partial pivoting) are required! The condition $S \succ 0$ automatically prevents numerical growth, guaranteeing that roundoff error remains bounded.

3.3 Invariance Under Inversion, Addition, and Congruence Transformations $M^{\mathsf{T}}SM$

Positive definite matrices form a convex cone with exceptional algebraic stability:

- **Closure Under Addition:** If $S_1 \succ 0$ and $S_2 \succ 0$, then $S_1 + S_2 \succ 0$. (Proof: $\mathbf{x}^{\mathsf{T}}(S_1 + S_2)\mathbf{x} = \mathbf{x}^{\mathsf{T}}S_1\mathbf{x} + \mathbf{x}^{\mathsf{T}}S_2\mathbf{x} > 0$).
- **Closure Under Inversion:** If $S \succ 0$, then $S^{-1} \succ 0$. Since $S = Q\,\Lambda\,Q^{\mathsf{T}}$, its inverse is $S^{-1} = Q\,\Lambda^{-1}\,Q^{\mathsf{T}}$. The eigenvalues of S^{-1} are $1/\lambda_i > 0$, and its eigenvectors are completely identical to those of S.
- **Congruence Invariance:** For any invertible matrix $M \in \mathbb{R}^{n \times n}$, $M^{\mathsf{T}}\,S\,M \succ 0$. For any $\mathbf{x} \neq \mathbf{0}$, set $\mathbf{y} = M\mathbf{x} \neq \mathbf{0}$; then $\mathbf{x}^{\mathsf{T}}(M^{\mathsf{T}}S\,M)\mathbf{x} = \mathbf{y}^{\mathsf{T}}S\mathbf{y} > 0$.

Part IV: Geometry of Ellipses & Hyper-Ellipsoids

4.1 Quadratic Level Sets $x^T S x = 1$ and Orthogonal Diagonalization

The geometric manifestation of positive definiteness appears in the level curves and level surfaces of the energy function $f(\mathbf{x}) = \mathbf{x}^T S \mathbf{x} = 1$.

By the Spectral Theorem, $S = Q \Lambda Q^T$ where $Q = [\mathbf{q}_1 \dots \mathbf{q}_n]$ is an orthogonal matrix of eigenvectors. Introducing the rotated coordinate system $\mathbf{x} = Q \mathbf{y}$ (or $\mathbf{y} = Q^T \mathbf{x}$), the equation of the level surface becomes:

$$\mathbf{x}^T S \mathbf{x} = (Q \mathbf{y})^T S (Q \mathbf{y}) = \mathbf{y}^T (Q^T S Q) \mathbf{y} = \mathbf{y}^T \Lambda \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2 = 1$$

Because all $\lambda_i > 0$, this equation is the canonical uncoupled equation of an **un-tilted hyper-ellipsoid** in the coordinate frame defined by $\mathbf{y}$!

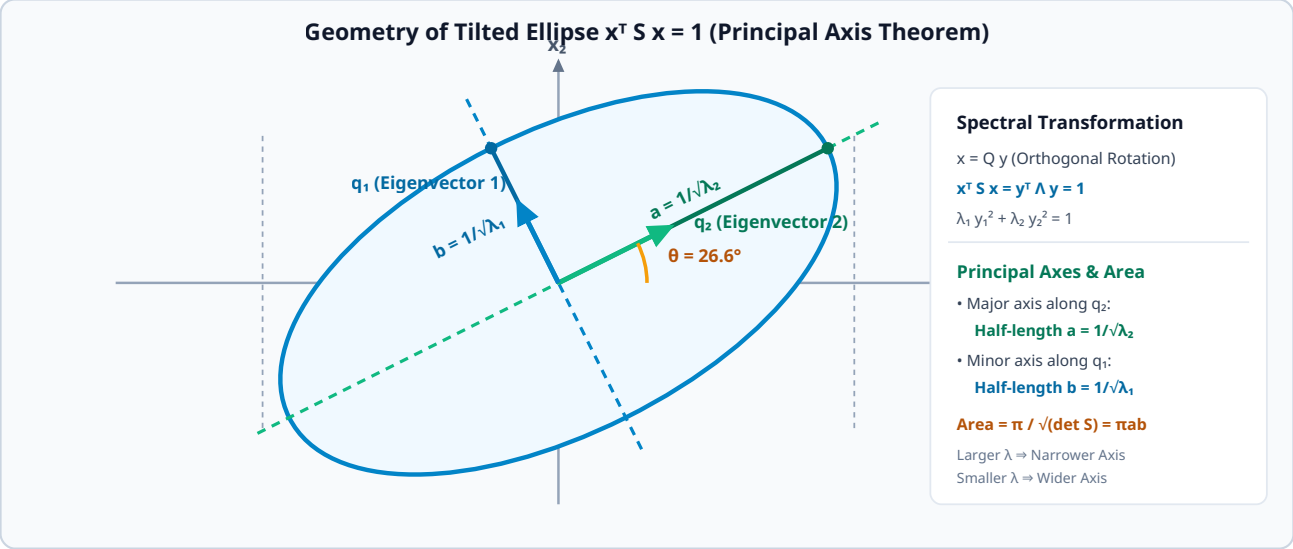


Figure 8.3: Metric Geometry of the Tilted Ellipse $x^T S x = 1$ — Principal Axes Align With Eigenvectors q_i With Semi-Axis Lengths $1/\sqrt{\lambda_i}$.

4.2 Principal Semi-Axes Alignment with Orthonormal Eigenvectors q_i

The transformation $\mathbf{x} = Q \mathbf{y}$ represents a pure rigid rotation (and possible reflection) of the coordinate axes:

- When $\mathbf{y} = [y_1, 0, \dots, 0]^T$ points along the first coordinate axis in $\mathbf{y}$-space, $\mathbf{x} = y_1 \mathbf{q}_1$ points directly along the eigenvector $\mathbf{q}_1$ in $\mathbf{x}$-space.
- Consequently, **the principal axes of the tilted ellipse point precisely along the orthonormal eigenvectors $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$!**

4.3 Semi-Axis Half-Lengths $a_i = 1/\sqrt{\lambda_i}$, Enclosed Area, and Hyper-Volume

To determine the length of the semi-axis along the i-th principal direction, consider the intercept along the y_i-axis where all other $y_j = 0$:

$$\lambda_i y_i^2 = 1 \Rightarrow y_i = \pm \frac{1}{\sqrt{\lambda_i}}$$

This establishes the celebrated **Inverse Square-Root Law** of quadratic forms:

Theorem 4.1 (Principal Semi-Axis Metric Theorem)

For any positive definite matrix $S \succ 0$, the quadratic level set $\mathbf{x}^T S\,\mathbf{x} = 1$ forms an un-tilted hyper-ellipsoid along the eigenvector frame with semi-axis lengths:

$$a_i = \frac{1}{\sqrt{\lambda_i}} \quad \text{for} \quad i = 1, 2, \ldots, n$$

Key Geometric Insight: The **largest eigenvalue** λ_{max} produces the **shortest semi-axis** (the direction of steepest energy rise and strongest confinement). The **smallest eigenvalue** λ_{min} produces the **longest semi-axis** (the direction of flattest energy rise).

In two dimensions $\mathbb{R}^2$, the enclosed area of the ellipse is given by:

$$\text{Area} = \pi\, a\, b = \pi \cdot \frac{1}{\sqrt{\lambda_1}} \cdot \frac{1}{\sqrt{\lambda_2}} = \frac{\pi}{\sqrt{(\lambda_1 \lambda_2)}} = \frac{\pi}{\sqrt{\det(S)}}$$

In n dimensions, the volume of the hyper-ellipsoid is obtained by scaling the unit sphere volume:

$$\text{Vol}(E) = \frac{V_n(\text{unit sphere})}{\sqrt{\det(S)}}$$

Part V: Minimum Principles & Advanced Applications

5.1 The Rayleigh Quotient R(x): Variational Stationarity and Spectral Bounds

For any non-zero vector $\mathbf{x} \neq \mathbf{0}$, the **Rayleigh quotient** normalizes the quadratic form by vector length squared:

$$R(\mathbf{x}) = \frac{\mathbf{x}^T S\,\mathbf{x}}{\mathbf{x}^T\mathbf{x}}$$

Theorem 5.1 (Stationary Property of the Rayleigh Quotient)

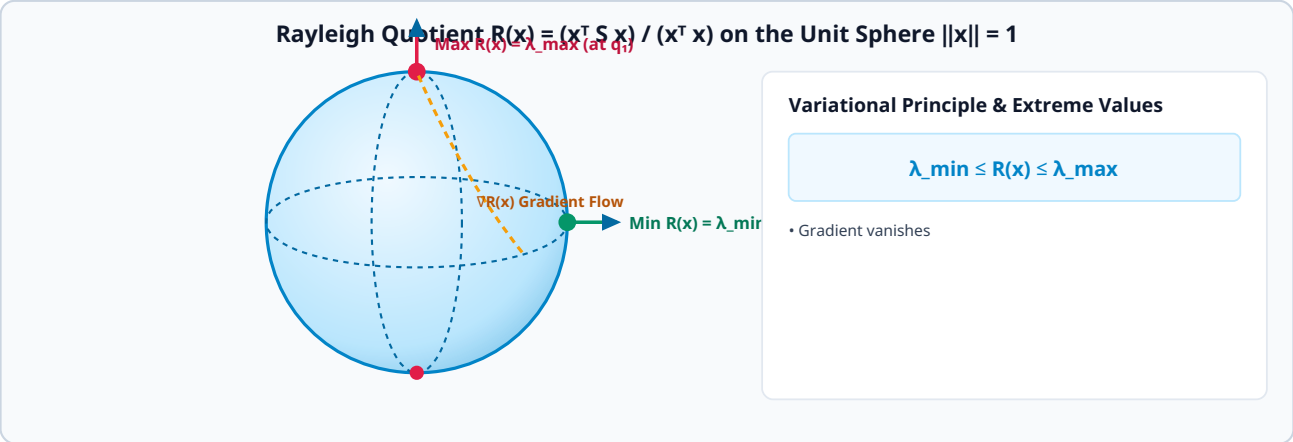
The gradient of the Rayleigh quotient vanishes if and only if $\mathbf{x}$ is an eigenvector of S:

$$\nabla R(\mathbf{x}) = \frac{2}{\mathbf{x}^T\mathbf{x}}\,[S\mathbf{x} - R(\mathbf{x})\,\mathbf{x}] = \mathbf{0} \;\Leftrightarrow\; S\mathbf{x} = R(\mathbf{x})\,\mathbf{x}$$

Consequently, the **critical values** of the Rayleigh quotient are precisely the eigenvalues λ_i, and the **critical points** are the corresponding eigenvectors $\mathbf{q}_i$!

Moreover, for all non-zero vectors $\mathbf{x} \neq \mathbf{0}$, the quotient is strictly bounded by the extreme eigenvalues:

$$\lambda_{\min} \leq \frac{\mathbf{x}^T S\,\mathbf{x}}{\mathbf{x}^T\mathbf{x}} \leq \lambda_{\max}$$



iff x is an eigenvector: ∇R(x) = 0 ⇔ Sx = R(x) x • Courant-Fischer Min-Max Theorem: Captures intermediate eigenvalues on orthogonal subspaces: λ_k = max min R(x) Connection to SVD: ||Ax|| / ||x|| achieves σ_1 = √(λ_max)

Figure 8.4: Variational Topography of the Rayleigh Quotient on the Unit Sphere ||x|| = 1 — Extreme Values Occur at Principal Eigenvectors.

5.2 Multivariable Second Derivative Test: The Hessian Matrix $H \succ 0$ in Optimization

Consider a smooth multivariable function $F: \mathbb{R}^n \to \mathbb{R}$. At a critical point $\mathbf{x}_0$ where the gradient vanishes $\nabla F(\mathbf{x}_0) = \mathbf{0}$, the multivariable Taylor expansion about $\mathbf{x}_0$ takes the form:

$$F(\mathbf{x}) = F(\mathbf{x}_0) + \nabla F(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) + \tfrac{1}{2} (\mathbf{x} - \mathbf{x}_0)^T H (\mathbf{x} - \mathbf{x}_0) + O(\|\mathbf{x} - \mathbf{x}_0\|^3)$$

where H is the **Hessian matrix** of second-order partial derivatives: $H_{ij} = \partial^2 F / \partial x_i \partial x_j$.

- If $H(\mathbf{x}_0) \succ 0$ (Positive Definite): $(\mathbf{x} - \mathbf{x}_0)^T H (\mathbf{x} - \mathbf{x}_0) > 0$ for all directions, guaranteeing a **strict local minimum**.
- If $H(\mathbf{x}_0) \prec 0$ (Negative Definite): $(\mathbf{x} - \mathbf{x}_0)^T H (\mathbf{x} - \mathbf{x}_0) < 0$, guaranteeing a **strict local maximum**.
- If $H(\mathbf{x}_0)$ is Indefinite: (eigenvalues of mixed signs), $\mathbf{x}_0$ is a **saddle point**.

5.3 Gramian Structure & The Bridge to the Singular Value Decomposition (A^TA and AA^T)

For any rectangular data matrix $A \in \mathbb{R}^{m \times n}$, the square matrices A^TA (size $n \times n$) and AA^T (size $m \times m$) are symmetric and positive semidefinite:

$$\mathbf{x}^T (A^TA)\, \mathbf{x} = \|A\mathbf{x}\|^2 \geq 0, \qquad \mathbf{y}^T (A\,A^T)\, \mathbf{y} = \|A^T\mathbf{y}\|^2 \geq 0$$

The non-zero eigenvalues of A^TA and $A\,A^T$ are identical and equal to the **squared singular values** σ_i^2 of A! The orthonormal eigenvectors $\mathbf{v}_i$ of A^TA form the right singular vectors, while the orthonormal eigenvectors $\mathbf{u}_i$ of $A\,A^T$ form the left singular vectors in the Singular Value Decomposition:

$$A = U \Sigma V^T = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

The matrix operator 2-norm $\|A\|_2 = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = \sigma_1 = \sqrt{\lambda_{\max}(A^TA)}$. Thus, positive definiteness and Rayleigh optimization serve as the mathematical engine powering the SVD.

Part VI: Fast-Track Hand Calculation Shortcut Compendium

⚡ FAST-TRACK 8.1: Instant 2×2 and 3×3 Sylvester Minor Tests

2×2 Mental Math Shortcut: For $S = \begin{vmatrix} a & b \\ b & c \end{vmatrix}$, test two numbers in 5 seconds:

1. $a > 0$
2. $ac - b^2 > 0$

Pitfall Guard: If $a \le 0$ or $c \le 0$, stop immediately — S can never be positive definite!

3×3 Shortcut: Evaluate only 3 determinants: $d_1 = s_{11} > 0$, $d_2 = s_{11}s_{22} - s_{12}^2 > 0$, and $\det(S) > 0$. Avoids solving the characteristic cubic polynomial!

⚡ FAST-TRACK 8.2: Gershgorin Strict Diagonal Dominance Instant Pass

The Rule: If a symmetric matrix with strictly positive diagonal entries is **strictly diagonally dominant**:

$s_{ii} > \sum_{j \neq i} |s_{ij}|$ for all $i = 1, \ldots, n$

then every eigenvalue $\lambda_i > 0$, and S is **guaranteed strictly positive definite** without computing a single pivot or determinant!

Example:

$\begin{vmatrix} 5 & 2 & 1 \\ 2 & 6 & 3 \\ 1 & 3 & 7 \end{vmatrix}$ has $5 > 3$, $6 > 5$, $7 > 4$. Strictly positive definite by inspection in 3 seconds!

⚡ FAST-TRACK 8.3: Direct Column-Wise Recursive Cholesky Extraction

The Rule: Skip LDL^T conversion and compute lower-triangular entries of L (where $S = LL^T$) column-by-column:

Diagonal: $L_{jj} = \sqrt{[s_{jj} - \sum_{k=1}^{j-1} L_{jk}^2]}$

Off-Diagonal ($i > j$): $L_{ij} = \frac{1}{L_{jj}} [s_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk}]$

Safety Guard: If any term under the square root is ≤ 0, stop: S is not positive definite.

⚡ FAST-TRACK 8.4: Instant Ellipse Metric Extraction: Tilt Angle & Area

The Rule: For the 2D ellipse $x^T S x = 1$ with $S = \begin{vmatrix} a & b \\ b & c \end{vmatrix}$:

Enclosed Area: Area $= \frac{\pi}{\sqrt{\det(S)}} = \frac{\pi}{\sqrt{(ac - b^2)}}$

Major Axis Tilt Angle: $\tan(2\theta) = \frac{2b}{a - c} \Rightarrow \theta = \frac{1}{2} \operatorname{atan2}(2b, a - c)$

Bypasses full eigenvector construction whenever only metric area and orientation are required.

⚡ FAST-TRACK 8.5: Quadratic Cost Function Minimization Shortcut

The Rule: For the general quadratic cost function $J(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T S\mathbf{x} - \mathbf{b}^T\mathbf{x} + c$ with $S \succ 0$:
Unique Minimizer: $\mathbf{x}^* = S^{-1}\mathbf{b}$
Absolute Minimum Value: $J(\mathbf{x}^*) = c - \frac{1}{2}\mathbf{b}^T S^{-1}\mathbf{b} = c - \frac{1}{2}\mathbf{b}^T\mathbf{x}^*$
Eliminates the need to compute multivariable partial derivatives by hand.

⚡ FAST-TRACK 8.6: Positive Semidefinite vs. Definite Nullspace Guard

The Rule: If $\det(S) = 0$ while all leading principal minors $\det(S_k) \geq 0$, S is **positive semidefinite** ($S \succeq 0$), not strictly positive definite!
Any non-zero vector in the nullspace $\mathbf{v} \in \text{null}(S)$ yields zero energy: $\mathbf{v}^T S\mathbf{v} = 0$. The energy surface is a flat trough along $\mathbf{v}$, meaning there is no unique minimum point.

Part VII: Step-by-Step Rigorous Worked Problems

Problem 8.1: Comprehensive 5-Way Equivalence Verification for Tridiagonal Matrix

5-Way Test Audit

PROBLEM STATEMENT

Consider the second-difference tridiagonal symmetric matrix S = $\begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix}$. Rigorously prove that S is strictly positive definite using all 5 equivalent tests.

TEST 1: THE ENERGY TEST (SUM OF SQUARES)

Expand $\mathbf{x}^T S \mathbf{x}$ explicitly:

$$\mathbf{x}^T S \mathbf{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 = x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2$$

As a sum of four real squares, $\mathbf{x}^T S \mathbf{x} \geq 0$. If $\mathbf{x}^T S \mathbf{x} = 0$, then $x_1 = 0 \Rightarrow x_1 - x_2 = 0 \Rightarrow x_2 = 0 \Rightarrow x_2 - x_3 = 0 \Rightarrow x_3 = 0$ $\Rightarrow \mathbf{x} = \mathbf{0}$. Thus $\mathbf{x}^T S \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. ✓

TEST 2: THE PIVOT TEST VIA GAUSSIAN ELIMINATION

Perform symmetric elimination without row swaps:

• Pivot 1: $d_1 = 2 > 0$. Multiplier $l_{21} = -1/2$.

• Row 2 update: $2 - (-1/2)(-1) = 2 - 1/2 = 3/2 \Rightarrow d_2 = 3/2 > 0$. Multiplier $l_{32} = -1 / (3/2) = -2/3$.

• Row 3 update: $2 - (-2/3)(-1) = 2 - 2/3 = 4/3 \Rightarrow d_3 = 4/3 > 0$.

All pivots (2, 3/2, 4/3) are strictly positive! ✓

TEST 3: SYLVESTER'S LEADING PRINCIPAL MINORS TEST

Compute upper-left subdeterminants:

• $\det(S_1) = |2| = 2 > 0$.

• $\det(S_2) = (2)(2) - (-1)^2 = 4 - 1 = 3 > 0$.

• $\det(S_3) = 2 \cdot \det(S_2) - (-1) \cdot [(-1)(2)] = 2(3) - 2 = 6 - 2 = 4 > 0$.

Notice pivots match minor ratios: $d_1 = 2$, $d_2 = 3/2$, $d_3 = 4/3$. All minors > 0! ✓

TEST 4: THE EIGENVALUE TEST

Characteristic polynomial: $\det(S - \lambda I) = (2 - \lambda)[(2 - \lambda)^2 - 2] = 0$.
Eigenvalues: $\lambda_1 = 2 - \sqrt{2} \approx 0.5858 > 0$, $\lambda_2 = 2.0 > 0$, $\lambda_3 = 2 + \sqrt{2} \approx 3.4142 > 0$. All $\lambda_i > 0$! ✓

TEST 5: GRAM MATRIX FACTORIZATION $S = A_0^T A_0$

Let A_0 be the 4 × 3 backward difference matrix: $A_0 =$

$$\begin{vmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{vmatrix}$$

. Direct multiplication yields $A_0^T A_0 = S$. The columns of A_0 are clearly linearly independent (rank = 3). Positive definiteness confirmed by all 5 tests! ✓

Problem 8.2: Parameter-Dependent Positive Definiteness Boundary Analysis S(c)

Parameter Analysis

PROBLEM STATEMENT

Determine the exact range of the real parameter c for which the matrix $S(c) =$

$$\begin{vmatrix} 2 & c & 1 \\ c & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix}$$

is strictly positive definite ($S \succ 0$). Identify where it becomes positive semidefinite.

STEP 1: LEADING PRINCIPAL MINOR DETERMINANTS

Apply Sylvester's Criterion to the three upper-left minors:

• **Minor 1:** $\det(S_1) = 2 > 0$. (Always strictly positive).

• **Minor 2:** $\det(S_2) = (2)(2) - c^2 = 4 - c^2$. Requiring $\det(S_2) > 0$ yields $c^2 < 4 \Rightarrow -2 < c < 2$.

• **Minor 3 (Full Determinant):** Expand $\det(S)$ along Row 3: $\det(S) = 1 \cdot (c - 2) - 1 \cdot (2 - c) + 2 \cdot (4 - c^2) = (c - 2) + (c - 2) - 2(c - 2)(c + 2) = (c - 2)[2 - 2(c + 2)] = (c - 2)(-2c - 2) = -2(c - 2)(c + 1) = 2(2 - c)(c + 1)$.

STEP 2: INTERVAL INTERSECTION

For $\det(S_3) > 0$, we require $(2 - c)(c + 1) > 0 \Rightarrow -1 < c < 2$.

Intersecting with Minor 2 ($-2 < c < 2$) gives: **Strict Positive Definiteness:** $-1 < c < 2$.

STEP 3: VERIFICATION VIA ELIMINATION PIVOTS & BOUNDARY ANALYSIS

Perform Gaussian elimination to verify pivots:

• $d_1 = 2 > 0$.

• $d_2 = \det(S_2)/d_1 = (4 - c^2)/2 > 0$ for $|c| < 2$.

• $d_3 = \det(S_3)/\det(S_2) = [2(2 - c)(c + 1)] / [(2 - c)(2 + c)] =$

$$\frac{2(c + 1)}{c + 2}.$$

For $d_3 > 0$ under $-2 < c < 2$, the denominator $c + 2 > 0$, so we must have $c + 1 > 0 \Rightarrow c > -1$.

Semidefinite Boundary: At $c = -1$, $\det(S) = 0$ and $d_3 = 0$ while $d_1 = 2$, $d_2 = 3/2$. $S(-1)$ is positive semidefinite with nullvector $\mathbf{v} = [1, 1, -1]^T$ (with explicit check: $S(-1)\mathbf{v} = \mathbf{0}$ and $\mathbf{v}^T S(-1)\mathbf{v} = 0$; note that $[1, -1, 0]^T$ is the nullvector at boundary $c = 2$).

Problem 8.3: Completing Squares, LDL^T, and Cholesky Factorization

Cholesky & Solving

PROBLEM STATEMENT

For the symmetric matrix S =

$$\begin{vmatrix} 4 & 6 & 2 \\ 6 & 13 & 5 \\ 2 & 5 & 6 \end{vmatrix}:$$

(a) Complete the square for the quadratic form $f(\mathbf{x}) = \mathbf{x}^T S\,\mathbf{x}$.

(b) Construct the Cholesky factor L_{chol} (such that $S = L_{chol}\,L_{chol}^T$).

(c) Solve the linear system $S\mathbf{x} = \mathbf{b}$ where $\mathbf{b} = [12, 28, 14]^T$ using forward and back substitution.

STEP 1: COMPLETING THE SQUARE & PIVOTS

Group terms systematically by variable:

$f(x, y, z) = 4x^2 + 12xy + 4xz + 13y^2 + 10yz + 6z^2$

$= 4[x^2 + 2x($

$\frac{3}{2}y + \frac{1}{2}z) + (\frac{3}{2}y + \frac{1}{2}z)^2] + 13y^2 + 10yz + 6z^2 - 4(\frac{9}{4}y^2 + \frac{3}{2}yz + \frac{1}{4}z^2)$

$= 4(x + \frac{3}{2}y + \frac{1}{2}z)^2 + 4y^2 + 4yz + 5z^2$

$= 4(x + \frac{3}{2}y + \frac{1}{2}z)^2 + 4(y + \frac{1}{2}z)^2 + 4z^2.$

The pivots are $d_1 = 4$, $d_2 = 4$, $d_3 = 4$. All strictly positive $\Rightarrow S \succ 0$!

STEP 2: CHOLESKY FACTOR L_CHOL = $L_0\,D^{1/2}$

From the multipliers: L_0 =

$$\begin{vmatrix} 1 & 0 & 0 \\ 3/2 & 1 & 0 \\ 1/2 & 1/2 & 1 \end{vmatrix} \text{ and } D^{1/2} = \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix}.$$

Multiplying gives: $L_{chol} = L_0\,D^{1/2} = \begin{vmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 1 & 2 \end{vmatrix}$.

Sanity Check: $L_{chol}\,L_{chol}^T = \begin{vmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 1 & 2 \end{vmatrix} \begin{vmatrix} 2 & 3 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{vmatrix} = \begin{vmatrix} 4 & 6 & 2 \\ 6 & 13 & 5 \\ 2 & 5 & 6 \end{vmatrix} = S$. Exact match!

STEP 3: TWO-STAGE FORWARD AND BACKWARD SUBSTITUTION

Solve $S\mathbf{x} = \mathbf{b} \Leftrightarrow L_{chol}\,\mathbf{c} = \mathbf{b}$, then $L_{chol}^T\,\mathbf{x} = \mathbf{c}$:

• Forward: $2c_1 = 12 \Rightarrow c_1 = 6$.

$3c_1 + 2c_2 = 28 \Rightarrow 18 + 2c_2 = 28 \Rightarrow c_2 = 5$.

$1c_1 + 1c_2 + 2c_3 = 14 \Rightarrow 6 + 5 + 2c_3 = 14 \Rightarrow c_3 = 1.5$.

• Back: $2x_3 = 1.5 \Rightarrow x_3 = 0.75$.

$2x_2 + 1x_3 = 5 \Rightarrow 2x_2 = 4.25 \Rightarrow x_2 = 2.125$.

$2x_1 + 3x_2 + 1x_3 = 6 \Rightarrow 2x_1 = 6 - 6.375 - 0.75 = -1.125 \Rightarrow x_1 = -0.5625$.

Problem 8.4: Complete Spectral and Metric Geometry of a Tilted Ellipse

Ellipse Geometry

PROBLEM STATEMENT

An elastic energy contour is defined by $5x_1^2 - 4x_1x_2 + 8x_2^2 = 36$.

(a) Express in canonical matrix form $\mathbf{x}^T S\,\mathbf{x} = 1$.
(b) Calculate eigenvalues and orthonormal eigenvectors of S.
(c) Determine the exact semi-axis lengths, tilt angle θ, and enclosed area.

STEP 1: CANONICAL MATRIX FORM

Divide the equation by 36:

$$\frac{5}{36}x_1^2 - \frac{4}{36}x_1x_2 + \frac{8}{36}x_2^2 = 1 \Rightarrow \mathbf{x}^T S\,\mathbf{x} = 1, \quad \text{where } S = \frac{1}{36}\begin{vmatrix} 5 & -2 \\ -2 & 8 \end{vmatrix}.$$

STEP 2: EIGENVALUES & SEMI-AXIS LENGTHS

Let $S_0 = \begin{vmatrix} 5 & -2 \\ -2 & 8 \end{vmatrix}$. $\mathrm{Tr}(S_0) = 13$, $\det(S_0) = 40 - 4 = 36$.

Roots: $\mu^2 - 13\mu + 36 = (\mu - 9)(\mu - 4) = 0 \Rightarrow \mu_1 = 9, \mu_2 = 4$.

Dividing by 36 gives eigenvalues of S: $\lambda_1 = 9/36 = 1/4$ and $\lambda_2 = 4/36 = 1/9$.

- **Major Semi-Axis Half-Length** (smaller λ): $a = 1/\sqrt{\lambda_2} = 1/\sqrt{(1/9)} = 3$.
- **Minor Semi-Axis Half-Length** (larger λ): $b = 1/\sqrt{\lambda_1} = 1/\sqrt{(1/4)} = 2$.

STEP 3: PRINCIPAL AXIS EIGENVECTORS & TILT ANGLE

For $\mu_2 = 4$ (Major Axis): $(S_0 - 4I)\mathbf{v} = \begin{vmatrix} 1 & -2 \\ -2 & 4 \end{vmatrix}\mathbf{v} = \mathbf{0} \Rightarrow v_1 - 2v_2 = 0 \Rightarrow \mathbf{q}_2 = \frac{1}{\sqrt{5}}\begin{vmatrix} 2 \\ 1 \end{vmatrix}$.

Tilt angle: $\tan\theta = 1/2 \Rightarrow \theta = \arctan(0.5) \approx 26.57°$.

For $\mu_1 = 9$ (Minor Axis): $(S_0 - 9I)\mathbf{v} = \begin{vmatrix} -4 & -2 \\ -2 & -1 \end{vmatrix}\mathbf{v} = \mathbf{0} \Rightarrow \mathbf{q}_1 = \frac{1}{\sqrt{5}}\begin{vmatrix} -1 \\ 2 \end{vmatrix}$.

Orthogonality check: $\mathbf{q}_1^T\mathbf{q}_2 = \frac{1}{5}(-2 + 2) = 0$. ✓

STEP 4: ENCLOSED METRIC AREA

Area $= \pi\,a\,b = \pi\,(3)(2) = 6\pi \approx 18.8496$.

Verification via Determinant Formula: $\pi / \sqrt{\det(S)} = \pi / \sqrt{(1/36)} = 6\pi$. Perfectly consistent! ✓

Problem 8.5: Multivariable Optimization via Hessian & Rayleigh Extremal Verification

Optimization & Rayleigh

PROBLEM STATEMENT

Part A: Find all critical points of $f(x, y) = x^3 + y^3 - 3xy$. Compute the Hessian matrix H at each critical point and classify using leading principal minors.
Part B: For the matrix $S = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}$, determine the maximum and minimum values of the Rayleigh quotient $R(\mathbf{x})$ and verify the bounds for the test vector $\mathbf{x} = [2, 1]^T$.

PART A: HESSIAN CRITICAL POINT CLASSIFICATION

Gradient: $\nabla f = [3x^2 - 3y, 3y^2 - 3x]^T = \mathbf{0} \Rightarrow y = x^2$ and $x = y^2 \Rightarrow x = x^4 \Rightarrow x(x^3 - 1) = 0$.
Real critical points: $P_0 = (0, 0)$ and $P_1 = (1, 1)$.

General Hessian matrix: $H(x, y) = \begin{vmatrix} 6x & -3 \\ -3 & 6y \end{vmatrix}$.

• At $P_0 = (0, 0)$: $H(0, 0) = \begin{vmatrix} 0 & -3 \\ -3 & 0 \end{vmatrix}$. $\det(H_1) = 0$, $\det(H_2) = -9 < 0$. Eigenvalues $\lambda = \pm 3$ (mixed signs). H is **Indefinite** $\Rightarrow$ (0, 0) is a **Saddle Point**.

• At $P_1 = (1, 1)$: $H(1, 1) = \begin{vmatrix} 6 & -3 \\ -3 & 6 \end{vmatrix}$. $\det(H_1) = 6 > 0$, $\det(H_2) = 36 - 9 = 27 > 0$. Both leading minors are positive!

H is **Positive Definite** ($H \succ 0$) $\Rightarrow$ (1, 1) is a **Strict Local Minimum** with value $f(1, 1) = -1$.

PART B: RAYLEIGH QUOTIENT BOUNDS VERIFICATION

For $S = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}$, $\mathrm{Tr}(S) = 6$, $\det(S) = 8$.

Eigenvalues: $\lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2) = 0 \Rightarrow \lambda_{max} = 4, \lambda_{min} = 2$.

Eigenvectors: $\mathbf{q}_1 = \frac{1}{\sqrt{2}}[1, 1]^T$ (achieving max $R = 4$) and $\mathbf{q}_2 = \frac{1}{\sqrt{2}}[1, -1]^T$ (achieving min $R = 2$).

Now evaluate for arbitrary test vector $\mathbf{x} = [2, 1]^T$:

• Numerator: $\mathbf{x}^T S \mathbf{x} = [2, 1] \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} \begin{vmatrix} 2 \\ 1 \end{vmatrix} = [2, 1] [7, 5]^T = 14 + 5 = 19$.

• Denominator: $\mathbf{x}^T \mathbf{x} = 2^2 + 1^2 = 5$.

• Quotient: $R(\mathbf{x}) = 19/5 = 3.8$.

Sanity check: $2 \le 3.8 \le 4$. The Rayleigh bound $\lambda_{min} \le R(\mathbf{x}) \le \lambda_{max}$ is strictly satisfied! ✓

Part VIII: Grand Master Summary & Matrix Taxonomy

8.1 Definitive Matrix Definiteness Property Taxonomy Table

The following master reference table synthesizes the five definiteness classes for real symmetric matrices:

Definiteness Class	Energy Sign $\mathbf{x}^T S \mathbf{x}$	Eigenvalues λ_i	Pivots d_i	Sylvester Minors $\det(S_k)$	Geometric Energy Surface
Positive Definite ($S \succ 0$)	> 0 for all $\mathbf{x} \neq \mathbf{0}$	All $\lambda_i > 0$	All $d_i > 0$	All $\det(S_k) > 0$	Upward Paraboloid Bowl (Strict Min at **0**)
Positive Semidefinite ($S \succeq 0$)	≥ 0 for all $\mathbf{x}$	All $\lambda_i \geq 0$	All $d_i \geq 0$	**All** principal minors ≥ 0	Parabolic Valley / Flat Trough
Negative Definite ($S \prec 0$)	< 0 for all $\mathbf{x} \neq \mathbf{0}$	All $\lambda_i < 0$	All $d_i < 0$	Signs alternate: $(-, +, -, +)$	Downward Inverted Dome (Strict Max at **0**)
Negative Semidefinite ($S \preceq 0$)	≤ 0 for all $\mathbf{x}$	All $\lambda_i \leq 0$	All $d_i \leq 0$	Non-positive / alternating signs	Inverted Valley / Flat Ridge
Indefinite	Positive & negative values	Mixed ($\lambda_+ > 0 > \lambda_-$)	Mixed (+ and −)	$\det(S_k)$ fails both patterns	Hyperbolic Saddle Surface (Pringles Chip)

⚡ **FAST-TRACK MASTER RECAP: 4 Golden Rules for Exams and Research**

1. **Rule 1 (The Pivot-Minor Hierarchy):** For 2×2 matrices, check $a > 0$ and $ac > b^2$. For 3×3, check $d_1, d_2, d_3 > 0$. Never solve the characteristic cubic polynomial unless eigenvalues are explicitly requested!
2. **Rule 2 (The Axis Inversion Principle):** In quadratic level sets $\mathbf{x}^T S \mathbf{x} = 1$, remember that semi-axis lengths scale as $a_i = 1/\sqrt{\lambda_i}$. The *largest* eigenvalue corresponds to the *narrowest* axis; the *smallest* eigenvalue corresponds to the *widest* axis.
3. **Rule 3 (The Sylvester Semidefinite Trap):** Never declare a matrix positive semidefinite by checking only leading principal minors! A zero leading minor does NOT mean all principal minors are non-negative. Check all principal minors or use the pivot test.
4. **Rule 4 (The Cholesky Efficiency Guard):** Whenever solving $S\mathbf{x} = \mathbf{b}$ for $S \succ 0$, always factor $S = LL^T$. It cuts operation count in half and is immune to numerical instability without row permutations.